36118 Applied Natural Language Processing

Master of Data Science and Innovation, UTS

Assessment Task 2B

End-to-End NLP Project — Technical Report

Sydney Liveability Explorer

Subject	36118 Applied Natural Language Processing
Submission	AT2 Part B — Group Technical Report
Due Date	Monday, 4 May 2026, 23:59
Suggested Length	10–15 pages (excluding appendix)
Group	Group 3, Autumn 2026
GitHub Repository	`https://github.com/Nelkit/sydney-liveability-ai`

Team Members

Last Name	First Name	Student ID	GitHub Username
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Contents

1 Executive Summary

Write this section last. Include one paragraph summarising each of the following: project objectives, data sources, NLP methods, findings, and outcomes.

2 Project Objectives and Scope

Describe the problem this project solves, the target user group, and what is in and out of scope for this submission.

3 Project Phases and Team Contributions

Describe project phases and map responsibilities to each team member. Each member must include an individual paragraph describing their specific contribution.

Phase Breakdown Table

Phase	Tasks	Team Member(s)
Phase 1: Data	Data acquisition and preprocessing from Reddit (Arc Shift), Community Insights PDF, BOCSAR CSV, TfNSW GTFS, OpenStreetMap (Overpass API), and City of Sydney ArcGIS.	
Phase 2: NLP Pipeline	Aspect-level sentiment analysis (DistilRoBERTa), sentence embeddings (MiniLM), ChromaDB ingestion, spaCy preprocessing.	
Phase 3: Multi-Agent Pipeline	CrewAI pipeline crew (pre-computation) and query crew (real-time), FastAPI backend, RAG with Claude API.	
Phase 4: Frontend	Next.js dashboard, Leaflet map, aspect radar chart, emotion profile visualisation, conversational chat UI.	
Phase 5: Evaluation and Report	NLP metrics, RAG quality, application evaluation, report, poster, peer review.	

4 Data Sources

4.1 Primary Data Sources

Document each source: name, access method, format, granularity, and contribution to the pipeline. Sources include: Community Insights PDF, Reddit via Arc Shift, BOCSAR CSV, TfNSW GTFS, OpenStreetMap via Overpass API, and City of Sydney ArcGIS.

4.2 Data Limitations and Ethical Considerations

Discuss Reddit as a public but self-selected source, privacy considerations, geographic bias toward inner Sydney suburbs, PDF granularity limitations, and the ethical use of community-generated content.

5 NLP Methods and Techniques

5.1 Text Preprocessing

Describe the preprocessing pipeline using spaCy: tokenisation, lemmatisation, and named entity recognition (NER) for suburb extraction. Also describe LangChain chunking strategy used for RAG ingestion.

5.2 Considered Approaches: Traditional NLP

Present a critical review of traditional NLP methods (VADER, LDA, TF-IDF) with reference to the literature. Explain why these approaches were considered but not adopted as the primary implementation for this project. Do not present them as implemented components.

5.3 Implemented Approaches: Modern NLP

5.3.1 Reddit Data Collection via Arc Shift

Explain why PRAW was replaced by Arc Shift for data collection. Report scale: 20,237 posts covering 563 suburbs.

5.3.2 Aspect-Level Sentiment Analysis (DistilRoBERTa)

Describe the eight sentiment dimensions analysed per suburb, the emotion profiling approach, and implementation by Kai (Ying-Kai Liao).

5.3.3 Sentence Embeddings (all-MiniLM-L6-v2) and ChromaDB

Describe the chunking strategy, embedding model, ChromaDB upsert process, and metadata fields (suburb, source) stored per chunk.

5.3.4 Multi-Agent Pipeline (CrewAI)

Describe both crews: the pipeline crew (pre-computes suburb data) and the query crew (responds in real time). Detail each agent, its role, tools, and how agents collaborate.

5.3.5 Retrieval-Augmented Generation (RAG + LLM)

Detail the RAG chain built with LangChain and ChromaDB. Describe how the Claude API (Sonnet) generates answers grounded in retrieved source chunks with citations.

5.4 Justification: Why Modern Over Traditional

Present an academic argument for each design choice: DistilRoBERTa over VADER, sentence embeddings over TF-IDF, RAG over keyword search, and CrewAI for modular orchestration. Cite Lewis et al. (2020), Reimers & Gurevych (2019), and Hutto & Gilbert (2014).

6 Findings and Evaluation

6.1 NLP Pipeline Results

Summarise aspect scores per suburb, emotion profiles, transport scores, POI density, and crime trends. Reference Appendix D for full tables and charts.

6.2 NLP Model Performance Metrics

Report evaluation metrics for DistilRoBERTa, retrieval quality for the RAG pipeline, and representative query-response pairs with citations from the LLM.

6.3 User-Facing Application Quality

Describe the dashboard, aspect radar visualisation, emotion profile chart, chat interface, and overall user experience quality based on walkthroughs or testing.

6.4 Temporal Trends

Summarise time-based patterns in Reddit sentiment per suburb where timestamps are available.

7 Outcomes and Value Added

Describe what the project achieved and the practical value delivered to the target user group.

8 Recommendations and Next Steps

List the highest-impact extensions: coverage of more suburbs, additional data sources, model fine-tuning, and integration with rental price APIs.

9 Conclusion

Summarise the project contributions, lessons learned, and future potential.

10 References

Use APA 7th edition for all citations. Key references include: DistilRoBERTa (Sanh et al., 2019), all-MiniLM-L6-v2 (Reimers & Gurevych, 2019), RAG (Lewis et al., 2020), CrewAI, and Arc Shift.

A Appendix A: System Architecture Diagram

Insert the full pipeline diagram: Arc Shift, GTFS, OSM, BOCSAR, and PDF ingestion feeding into DistilRoBERTa and ChromaDB, then CrewAI, RAG, and the frontend.

B Appendix B: Dashboard Screenshots

Insert annotated screenshots of the aspect radar, emotion profile, Leaflet map, and chat interface.

C Appendix C: Sample Code

Include representative code for `ingest_sentiment.py`, CrewAI agents, the RAG chain, and FastAPI endpoints.

D Appendix D: Evaluation Tables and Charts

Include suburb-level scores per sentiment dimension, emotion distributions, transport scores, and POI density charts.

E Appendix E: Traditional vs. Modern NLP Comparison Table

Table 2: Traditional vs. Modern NLP Approach Comparison

Approach	Type	Used	Rationale
VADER sentiment	Traditional	No	Lexicon-based, no aspect granularity
LDA topic modelling	Traditional	No	Unsupervised, low coherence on Reddit text
TF-IDF keyword extraction	Traditional	No	Frequency-based, no semantic understanding
DistilRoBERTa aspect sentiment	Modern	Yes	8-dimension aspect scoring, contextual
all-MiniLM-L6-v2 embeddings	Modern	Yes	Semantic similarity, compact and fast
RAG with Claude Sonnet	Modern	Yes	Grounded, cited answers from retrieved chunks
CrewAI multi-agent pipeline	Modern	Yes	Modular orchestration, scalable per suburb